

# Faculty of Engineering and Technology

The Faculty of Engineering and Technology (FET) is one of Indonesia's leading schools of engineering and technology recognized for the quality of its programs, faculty and facilities. We offer a premier educational experience for success in competitive careers in computer science, information technology, and engineering. According to industry experts, our approach makes our graduates the most competitive in Indonesia.

## COMPUTER SCIENCE PROGRAM

Sarjana Komputer Sampoerna University

Bachelor of Science in Electrical & Computer Engineering University of Arizona\*

Computer Science program offers a comprehensive foundation that permits its graduates to adapt to new technologies and new ideas in computing. Computer science spans a wide range, from theoretical and algorithmic foundations to developments in new areas; from theory to applications programming. Computer science graduates design, develop, and analyze various types of computer system from system infrastructure (hardware, operating systems, communication networks, etc.) to application technologies (mobile and web applications, databases, etc.).

### Competencies to achieve:

- Algorithms
- Computer Architecture and Organization
- Computational Science
- Discrete Structures
- Information Security
- Intelligent Systems
- Networking and Communications
- Operating Systems
- Programming
- Software Development
- Software Engineering

### Career Pathways

Our graduates of the Computer Science/ Informatics Study Program have many possible career pathways, which include but are not limited to:

- Software/ applications developers
- Game developers
- Computer systems analysts
- Computer systems engineers
- IoT engineers
- Computer network engineers
- Computer network administrators
- Database administrators
- Artificial Intelligence analysts / engineers
- Web systems developers
- Web administrators
- Software engineers

| SEMESTER I |                                                   |         |
|------------|---------------------------------------------------|---------|
| CODES      | COURSE TITLE                                      | CREDITS |
| GCOM1304   | Composition I (English)                           | 3       |
| GENG1301   | Introduction to Engineering                       | 3       |
| GSCI1303   | General Chemistry I (GE EP-Natural Scientist)     | 3       |
| GSCI1303L  | General Chemistry I Lab (GE EP-Natural Scientist) | 1       |
| GMAT1313   | Calculus I                                        | 3       |
| GSOC2304   | Social Problems (GE EP-Social Scientist)          | 3       |
| GHUM1207   | Kewarganegaraan                                   | 2       |
| Total      |                                                   | 18      |

| SEMESTER II |                                          |         |
|-------------|------------------------------------------|---------|
| CODES       | COURSE TITLE                             | CREDITS |
| GCOM1305    | Composition II (English)                 | 3       |
| GMAT1314    | Calculus II                              | 3       |
| COMP1401    | Computer Programming                     | 4       |
| GSCI1312    | General Physics with Calculus I          | 3       |
| GSCI1112L   | General Physics with Calculus I Lab      | 1       |
| GHUM1303    | World Religions (GE EP-Humanist)         | 3       |
| GHUM1301    | Pancasila (GE EP-Building Connections 1) | 3       |
| Total       |                                          | 19      |

| SEMESTER III |                                      |         |
|--------------|--------------------------------------|---------|
| CODES        | COURSE TITLE                         | CREDITS |
| GMAT2415     | Calculus III                         | 4       |
| GSCI2313     | General Physics with Calculus II     | 3       |
| GSCI2113L    | General Physics with Calculus II Lab | 1       |
| COMP2424     | Data Structures & Algorithms         | 4       |
| COMP2303     | Digital Systems                      | 3       |
| MTED3314     | Discrete Mathematics                 | 3       |
| HED 102A     | Succeeding as a Global Wildcat (UA)  | 0       |
| GHUM1208     | Indonesian Language                  | 2       |
| Total        |                                      | 20      |

| SEMESTER IV |                                                   |         |
|-------------|---------------------------------------------------|---------|
| CODES       | COURSE TITLE                                      | CREDITS |
| COMP1402    | Object Oriented Programming                       | 4       |
| GSCI2210    | Introduction to Optics and Thermodynamics         | 2       |
| COMP2304    | Computer Organization & Architecture              | 3       |
| GMAT2308    | Introduction to Ordinary Differential Equations   | 3       |
| VCDD1326    | Design Thinking & Theory (GE EP-Artist)           | 3       |
| GSOC2303    | General Psychology (GE EP-Building Connections 3) | 3       |
| HED 102B    | Succeeding as a Global Wildcat II (UA)            | 0       |
| Total       |                                                   | 18      |

| SEMESTER V |                                          |         |
|------------|------------------------------------------|---------|
| CODES      | COURSE TITLE                             | CREDITS |
| COMP3423   | Mathematics for Engineering Applications | 4       |
| COMP3408   | Database Management Systems              | 4       |
| ENGR3303   | Environmental Ethics                     | 3       |
| COMP4322   | Mobile Applications Development          | 3       |
| COMP3524   | Basic Circuits                           | 5       |
|            |                                          |         |
|            |                                          |         |
|            |                                          |         |
|            |                                          |         |
|            |                                          |         |
| Total      |                                          | 19      |

| SEMESTER VI |                                                  |         |
|-------------|--------------------------------------------------|---------|
| CODES       | COURSE TITLE                                     | CREDITS |
| COMP3309    | Computer Networks                                | 3       |
| COMP4429    | Microprocessor Organization (with Lab)           | 4       |
| COMP3326    | Circuit Theory                                   | 3       |
| GSOC1303    | Principles of Macroeconomics                     | 3       |
| COMP3312    | Human- Computer Interaction (incl. UI/UX Design) | 3       |
| COMP2306    | Elective:<br>Operating Systems                   | 3       |
| COMP3313    | Web Programming                                  |         |
| INFS2318    | Technopreneurship                                |         |
| Total       |                                                  | 19      |

| SEMESTER VII |                                 |         |
|--------------|---------------------------------|---------|
| CODES        | COURSE TITLE                    | CREDITS |
| STEM4301     | Internship for Computer Science | 3       |
| STEM4302     | Senior Capstone I               | 3       |
| COMP3311     | Software Engineering            | 3       |
| COMP4427     | Electronic Circuits (with lab)  | 4       |
| COMP4328     | Intro To Communications         | 3       |
| COMP3325     | Computer Architecture & Design  | 3       |
| Total        |                                 | 19      |

| SEMESTER VIII |                                                |         |
|---------------|------------------------------------------------|---------|
| CODES         | COURSE TITLE                                   | CREDITS |
| STEM4303      | Senior Capstone II                             | 3       |
| COMP3315      | Artificial Intelligence                        | 3       |
| COMP4330      | Knowledge System Engineering                   | 3       |
| COMP4331      | Fundamentals of Information & Network Security | 3       |
| COMP4332      | Computer Aided Logic Design                    | 3       |
| Total         |                                                | 15      |

|                                       |     |
|---------------------------------------|-----|
| TOTAL CREDIT HOURS FOR DEGREE PROGRAM | 148 |
|---------------------------------------|-----|